AI JOB TRENDS

Growth, Salary & Automation Risk (2024–2030)

Sector: Future of Work / Labor Market Analytics / AI & Automation

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1. Executive Summary

This project analyzes how AI adoption and automation are reshaping job markets across industries, focusing on job growth projections to 2030, salary drivers, and automation risk. Using a structured dataset of job roles and industry attributes, we built an end-to-end Python pipeline to clean data, compute business KPIs, perform exploratory data analysis (EDA), and run statistical tests and regression models.

What We Did

We organized the analysis into four core themes:

- Job Growth & Future Demand — distribution of growth rates, industry-level growth, and identification of top growing/declining jobs.
- Salary Drivers & AI Impact — salary distribution, salary differences by industry and AI impact, and the role of experience, education, and remote work.
- Automation Risk — distribution of automation risk and its relationship with job decline and growth.
- Workforce Structure — education demand, experience distribution, and remote-work patterns.

Key Findings (High-Level)

- Roles with higher AI exposure tend to show stronger growth potential, signaling a workforce shift toward AI-centric skills.
- Salaries are strongly shaped by industry, experience, education, and AI impact, with clear stratification across sectors.
- Automation risk is associated with job vulnerability: roles more exposed to automation are more likely to be declining.
- Remote-enabled roles often show higher salary patterns, especially in tech-driven domains.

Business Value

The results help decision-makers — employers, policy teams, and educators — prioritize where to invest in hiring (high-growth industries/roles), which roles need reskilling pathways (high-risk declining roles), and what drives compensation (experience, education, remote work, AI impact).

Deliverables

- A cleaned and aggregated dataset ready for Tableau visualization.
- KPI framework covering jobs, salary, growth, risk, and status distributions.
- EDA with statistical validation: correlation, hypothesis tests, regressions.
- Dashboard blueprint across four themes: growth, salary, risk, and workforce.

2. Sector Context & Problem Statement

2.1 Sector Context: AI, Automation, and Labor Markets

Organizations face growing uncertainty in workforce planning as AI tools and automation technologies rapidly change the employment landscape. Key challenges include:

- Which industries will expand employment demand through new job creation.
- Which job families will shrink due to task automation and process digitization.
- How compensation will evolve, particularly around premiums for AI-related skills.
- How education and experience translate into employability and pay.
- Which roles are most exposed to automation risk.

This makes workforce strategy increasingly data-driven. Hiring plans, training budgets, university curricula, and policy interventions need evidence rather than assumptions.

2.2 Problem Statement

This project addresses the following decision problem: How can we quantify and explain job growth, salary patterns, and job vulnerability (automation risk) across industries, and determine whether AI impact measurably affects growth and pay?

2.3 Business Questions

Growth & Demand

- Which industries are projected to grow the most by 2030?
- Which job roles are declining?
- How does AI impact relate to growth?

Salary

- Do AI-heavy roles pay more?
- How does experience affect salary?
- Which industries offer the highest pay?

Risk

- Which jobs are most at risk from automation?
- Do high-risk jobs decline more?
- Which industries are comparatively safer?

Workforce

- What education levels are most valuable?
- Does remote work influence salary and growth?
- How does experience affect growth potential?

3. Data Description

3.1 Data Source

Dataset: AI Job Trends / Future-of-Work job role dataset as provided for this project. The dataset captures projected job attributes from a 2024 baseline through 2030, covering role-level information on compensation, AI exposure, automation vulnerability, and workforce characteristics.

3.2 Dataset Structure

Column	Description	Type	Example
Job Title	Role name	String	ML Engineer
Industry	Industry category	String	Healthcare
Growth Rate	Projected job growth (0–1)	Float	0.32
Job Status	Trend direction of role demand	Categorical	Increasing
AI Impact	Impact level of AI on the role	Categorical	High
Salary	Annual salary (USD)	Integer	95,000
Experience Level	Years of experience required	Integer	4
Education Level	Required education level	Categorical	Master's
Automation Risk	Risk score of automation exposure (0–1)	Float	0.65
Remote Ratio	Remote work proportion (0–1)	Float	0.60

3.3 Derived Fields (Feature Engineering)

Derived Field	Description	Logic
Growth Category	Categorizes projected growth rate	High > 0.2, Medium 0–0.2, Low < 0
Risk Category	Automation risk segmentation	High > 0.7, Medium 0.4–0.7, Low < 0.4
Salary Range	Bucketed salary bands	Low / Medium / High based on quartiles
Experience Group	Grouped experience for dashboard slicing	0–2, 3–5, 6–10, 10+

3.4 Key Limitations

- Projection uncertainty: Growth rates to 2030 are forecasts and depend on underlying macroeconomic assumptions.
- Measurement ambiguity: AI impact and automation risk are modeled scores; definitions may differ across sources.
- Aggregation bias: Industry averages can hide within-industry variability across job families.
- Missing data: Salary, experience, and remote ratio may have inconsistent coverage across industries.
- Causality limitation: Observational data supports correlation and model-based explanation, not definitive causation.

4. Data Cleaning & ETL Methodology

The pipeline moves from raw data through cleaning, feature engineering, and aggregation to produce Tableau-ready outputs.

Step 1 — Ingest

- Load raw dataset (CSV/Excel) into pandas.
- Standardize column names to snake_case and enforce correct data types.

Step 2 — Cleaning

- Remove rows with missing identifiers (Job Title, Industry).
- Impute or drop missing numeric fields (Salary, Growth Rate, Automation Risk) based on impact assessment.
- Remove exact duplicates and duplicates by key columns.
- Validate ranges: growth rate within bounds, salary non-negative, remote ratio 0–1, risk score within defined scale.

Step 3 — Feature Engineering

- Create Risk Category (Low/Medium/High) from automation risk thresholds.
- Create AI Impact group (Low/Medium/High) from AI impact scores.
- Create Experience bands (0–2 / 3–5 / 6–10 / 10+) for segmentation.
- Normalize industry names (e.g., 'IT' and 'Technology' unified to one label).
- Ensure consistent job status labels across the dataset.

Step 4 — Aggregation for KPIs

- Industry-level: mean/median growth rate, mean salary, mean automation risk, share of declining/increasing roles.
- Job-level: top growing jobs, top declining jobs, top paying jobs, highest-risk jobs.
- Workforce: education distribution, experience distribution, remote ratio by industry.

Step 5 — Output for Tableau

- Export clean row-level table for flexible filtering.
- Export aggregated tables: industry summary, job summary, risk segmentation, workforce summary.
- Ensure Tableau-friendly types: numbers as numeric, categories as strings.

Quality Checks

- Row count before vs. after cleaning (documented in notebook outputs).
- Missing-value report per column.
- Summary statistics sanity check (min/median/max).
- Category counts for Industry, Job Status, AI Impact, and Risk Category.

5. KPI & Metric Framework

5.1 Core KPI Cards

KPI	Description	Formula
Total Jobs	Number of job records in the dataset	COUNT(Job Title)
Average Salary	Mean annual salary (USD)	AVG(Salary)
Average Growth Rate	Mean projected growth rate	AVG(Growth Rate)
Average Automation Risk	Mean automation risk score	AVG(Automation Risk)
% Increasing Jobs	Share of jobs in 'Increasing' status	COUNT(Increasing) / Total
% High Risk Jobs	Share of jobs in 'High' risk category	COUNT(High Risk) / Total

5.2 Metric Definitions

Metric	Type	Definition	Why It Matters
Growth Rate	Numeric	Projected change in demand by 2030	Signals future hiring needs
Salary	Numeric	Compensation level per role	Indicates skill premium
Automation Risk	Numeric	Exposure of tasks to automation	Indicates vulnerability
AI Impact	Cat/Numeric	Degree of AI relevance or exposure	Explains growth/salary shifts
Remote Ratio	Numeric	Share of remote feasibility/work	Linked to pay and talent access
Education Level	Categorical	Highest education requirement	Hiring + curriculum planning
Experience	Numeric/Cat	Required years/level	A major salary driver

5.3 KPI Mapping to Dashboards

Dashboard	Primary KPIs
Dashboard 1 — Job Growth & Future Demand	Total Jobs | Avg Growth Rate | % Increasing | % Declining
Dashboard 2 — Salary Drivers & AI Impact	Avg Salary | Max Salary | Salary Range
Dashboard 3 — Automation Risk & Vulnerability	Avg Risk | % High Risk Jobs

Dashboard	Primary KPIs
Dashboard 4 — Workforce Trends & Hiring Strategy	Avg Experience | Avg Remote Ratio

6. Exploratory Data Analysis (EDA)

6.1 Job Growth & Future Demand

Visuals Included

- Growth rate distribution (Histogram)
- Industry vs. average growth rate (Bar chart)
- Top 10 growing jobs (Bar)
- Top 10 declining jobs (Bar)
- Job status distribution (Stacked bar)
- AI impact vs. growth (Scatter) and Growth by AI group (Boxplot)

Written Insights

Growth is not uniform across roles. The growth rate histogram shows a wide spread, with a small cluster of roles exhibiting very strong projected expansion while others are flat or negative. Demand is likely to concentrate in specific job families rather than rise evenly across the economy.

Industry differences are pronounced. The industry-level bar chart highlights sectors with consistently higher average growth rates — these represent priority areas for hiring investment and workforce expansion planning.

Declining roles are structurally different from growing roles. The top declining jobs list is dominated by roles that are routine, clerical, or repetitive. In contrast, top growing roles include analytical, digital, AI-adjacent, or care-oriented work.

AI impact is positively associated with growth. Higher AI impact categories tend to have higher growth distributions, supporting the narrative that AI is creating complementary roles alongside automation.

6.2 Salary Analysis

Visuals Included

- Salary distribution (Histogram)
- Salary by industry (Boxplot)
- AI impact vs. salary (Bar)
- Experience vs. salary (Scatter/Line)
- Salary by education (Boxplot)
- Remote work ratio vs. salary (Scatter)

Written Insights

Salary distribution is typically right-skewed. Most roles cluster around a mid-range salary with a long tail of high-paying roles. Median salary is often a better 'typical pay' metric than the mean, which is pulled upward by outliers.

Pay differs significantly across industries. Industry boxplots show clear separation, reflecting differences in value creation, productivity, and skill premiums across sectors.

AI impact roles show a salary premium. Higher AI impact categories tend to command higher average salaries, consistent with market behavior where scarce AI-related skills attract compensation premiums.

Experience raises salary, but not always linearly. Salary tends to increase with experience, with sharper jumps after mid-level seniority and potential plateauing at senior levels.

Education is associated with higher salary bands. Higher median salaries are observed for advanced degrees (Master's/PhD), suggesting education is a meaningful earnings differentiator.

Remote-enabled roles can correlate with higher pay, especially in tech-oriented industries — though this is influenced by role type and industry mix.

6.3 Automation Risk

Visuals Included

- Risk distribution (Histogram)
- Top risk jobs (Bar)
- Industry vs. risk (Bar)
- Risk vs. job status (Boxplot)
- Risk vs. growth (Scatter)
- Risk category distribution (Bar/Pie)

Written Insights

Automation risk varies widely across roles, indicating not all jobs are equally exposed. The distribution typically shows a broad spread across the risk scale.

High-risk roles cluster around routine tasks. The top risk jobs chart features roles with standardized, repetitive workflows — providing a concrete target list for reskilling programs.

Risk differs by industry. Certain industries exhibit higher average exposure, indicating sector-level workforce vulnerability that requires targeted intervention.

Risk and job status relate meaningfully. Boxplots indicate that declining roles have higher risk distributions, consistent with automation-driven displacement patterns.

6.4 Workforce Structure (Education, Experience, Remote Work)

Visuals Included

- Education distribution (Bar)
- Experience distribution (Histogram)
- Remote ratio by industry (Bar)
- Experience vs. salary (Scatter)
- Experience vs. growth (Scatter)
- Remote work vs. growth (Boxplot)

Written Insights

Education demand is uneven. Labor market demand is concentrated around a few education levels, providing signals for curriculum planning and candidate pipeline development.

Experience is a strong differentiator. The relationship between experience and both salary and growth highlights that workforce planning must account for seniority mix — not just headcount.

Remote work is industry-dependent. Remote feasibility is shaped by the nature of work in each sector and is not universal.

Remote feasibility may influence outcomes. Remote vs. salary and remote vs. growth views help identify whether remote-enabled roles are associated with higher pay or stronger growth trajectories.

7. Statistical Analysis Results

7.1 Correlation Analysis

Method: Pearson/Spearman correlation across Salary, Growth Rate, Automation Risk, AI Impact, Experience, and Remote Ratio.

Purpose: Identify linear associations and potential predictors for regression modeling.

- Salary shows a positive association with Experience and AI Impact, indicating skill intensity and seniority are key drivers.
- Growth shows a positive association with AI Impact and a negative association with Automation Risk, suggesting higher-risk roles are less future-proof.
- Automation Risk shows a negative association with Job Status/Growth, aligning with job vulnerability patterns.

7.2 Hypothesis Test 1 — AI Impact vs. Growth

Question: Do high AI impact roles grow more than low AI impact roles?

Test: Two-sample t-test (or Mann–Whitney if non-normal). H0: Mean growth (High AI) = Mean growth (Low AI); H1: Mean growth differs between groups.

Result: If $p < 0.05$, AI impact is a statistically meaningful differentiator of growth; otherwise the observed difference may not be reliable beyond the sample. Exact values are drawn from notebook output.

7.3 Hypothesis Test 2 — AI Impact vs. Salary

Question: Do AI-heavy roles pay more?

Test: Two-sample t-test / Mann–Whitney.

Business Interpretation: A significant result confirms that AI skill exposure commands a measurable wage premium. An insignificant result suggests the premium may be industry-driven (confounded) rather than AI-specific.

7.4 Hypothesis Test 3 — Risk vs. Job Status

Question: Are high-risk jobs more likely to be declining?

Test: t-test comparing automation risk scores between declining vs. non-declining roles; or ANOVA across multiple job status groups.

Result: A statistically significant difference in average risk between declining and non-declining roles supports the hypothesis that automation risk predicts job vulnerability.

7.5 Hypothesis Test 4 — Education vs. Salary

Question: Does education level affect salary?

Test: ANOVA (or Kruskal–Wallis for non-normal distributions). Post-hoc Tukey HSD is applied where applicable to identify which education-level pairs differ significantly (e.g., Bachelor's vs. Master's).

7.6 Regression Model 1 — Salary Drivers

Model: Salary ~ Experience + AI Impact + Education + Remote Ratio (+ Industry controls)

Purpose: Quantify salary drivers while controlling for other variables.

- R^2 / Adjusted R^2: reported from notebook output.
- Experience coefficient: positive — each additional year of experience increases expected salary, holding other variables constant.
- AI Impact category coefficient: higher impact categories show a salary premium relative to the baseline.
- Remote ratio: remote-friendly roles show a measurable salary difference after controls.
- Industry controls: certain industries systematically pay more even after controlling for experience and AI impact.

7.7 Regression Model 2 — Growth Drivers

Model: Growth Rate ~ AI Impact + Automation Risk + Industry (+ Education/Remote if included)

- R^2: reported from notebook output.
- AI Impact coefficient: positive — higher AI exposure is associated with stronger growth.
- Automation Risk coefficient: typically negative — higher risk is associated with weaker growth.
- Industry dummies capture sector-level structural differences in growth trajectories.

Model limitations: multicollinearity between predictors, categorical encoding assumptions, potential non-linearity, and the observational (non-causal) nature of the data.

8. Dashboard Design (Tableau)

Dashboard 1 — Job Growth & Future Demand

Goal: Identify growth industries, expanding roles, and declining roles; evaluate AI impact on growth.

Element	Detail
KPI Cards	Total Jobs | Avg Growth | % Increasing | % Declining
Industry vs. Growth (Bar)	Which industries lead job growth?
Top 10 Growing Jobs (Bar)	Where hiring demand concentrates
Top 10 Declining Jobs (Bar)	Where displacement/reskilling is needed
AI Impact vs. Growth (Scatter)	Whether AI exposure aligns with growth
Growth Distribution (Histogram)	Overall market spread of growth rates
Active Filters	Industry | AI Impact | Job Status

Dashboard 2 — Salary Drivers & AI Impact

Goal: Explain salary distribution and quantify how AI impact, experience, and education relate to pay.

Element	Detail
KPI Cards	Avg Salary | Max Salary | Salary Range
Salary by Industry (Boxplot)	Which industries pay higher and how variable pay is
AI Impact vs. Salary (Bar)	Wage premium by AI exposure group
Top 10 Paying Jobs (Bar)	Highest-paying roles
Experience vs. Salary (Scatter)	Seniority-pay relationship
Salary Distribution (Histogram)	Skew and outliers in compensation
Active Filters	Industry | Education | AI Impact

Dashboard 3 — Automation Risk & Job Vulnerability

Goal: Identify jobs and industries at risk; test whether risky jobs decline more.

Element	Detail
KPI Cards	Avg Automation Risk | % High Risk Jobs
Industry vs. Risk (Bar)	Which sectors are most exposed
Top 10 High-Risk Jobs (Bar)	Vulnerable roles list
Safe Jobs (Bar/List)	Low-risk roles for stability planning
Risk vs. Growth (Scatter)	Whether risk predicts weaker growth

Element	Detail
Risk Distribution (Histogram)	Overall risk spread
AI Impact vs. Risk (Scatter)	Interaction between AI exposure and risk
Active Filters	Industry | Risk Category

Dashboard 4 — Workforce Trends & Hiring Strategy

Goal: Understand education, experience, and remote structure; derive hiring strategy implications.

Element	Detail
KPI Cards	Avg Experience | Avg Remote Ratio
Education Distribution (Bar)	What education levels dominate demand
Salary by Education (Boxplot)	Education–pay relationship
Growth by Education (Bar)	Education–growth relationship
Experience vs. Salary (Scatter)	Seniority-pay pattern
Remote vs. Salary (Scatter)	Remote-pay pattern
Remote vs. Growth (Boxplot)	Remote feasibility vs. growth
Active Filters	Education | Experience Level | Remote Ratio

9. Key Insights (Decision Language)

1. AI-exposed roles are consistently positioned in higher-growth segments, indicating workforce demand is shifting toward AI-adjacent capabilities by 2030.
2. Technology and Healthcare emerge as top growth industries — these sectors should be prioritized for talent expansion and recruiting investment.
3. Clerical and repetitive job roles show persistent decline, making them the highest-priority targets for reskilling and transition pathways.
4. High AI impact roles command a measurable salary premium, reinforcing AI skills as a compensation driver across industries.
5. Salary increases with experience but shows non-linear gains — stronger jumps appear after mid-level experience, with diminishing returns at senior levels.
6. Advanced education (Master's/PhD) aligns with higher salary bands, indicating that specialized education improves earning potential in the dataset.
7. Automation risk is concentrated in a subset of roles, enabling focused interventions rather than broad, unfocused workforce programs.
8. High-risk roles are disproportionately represented among declining jobs, supporting the hypothesis that automation exposure translates into labor market vulnerability.
9. Manufacturing shows higher automation exposure than most sectors, implying greater disruption risk and a need for technology-enabled upskilling initiatives.
10. Creative and strategic roles remain comparatively safer — they exhibit resilience where human judgment, communication, and design complexity dominate.
11. Remote-enabled roles tend to sit in higher salary ranges, especially within tech-driven industries, suggesting remote feasibility can improve talent access and pay competitiveness.
12. Industry effects remain strong even after accounting for experience and AI impact — compensation and growth planning should be sector-specific rather than one-size-fits-all.

10. Actionable Business Recommendations

Recommendation 1 — Prioritize Hiring in High-Growth Industries

Allocate recruiting budgets and learning programs toward roles and skills appearing in the top growth segments, particularly Technology and Healthcare. Focus on roles with both high AI impact and strong growth projections to maximize return on talent investment.

Recommendation 2 — Launch Targeted Reskilling Programs for High-Risk Roles

Focus transition programs on workers in routine clerical and repetitive jobs, moving them toward adjacent growth roles such as analytics, operations technology, customer success, and AI-support functions where demand is increasing and automation risk is lower.

Recommendation 3 — Introduce AI Skill Premiums and Structured Career Pathways

Formalize skill ladders — through certifications and internal mobility tracks — to retain AI-capable talent and attract candidates with relevant competencies. Where AI impact is tied to salary and growth, structured career pathways create a competitive retention advantage.

Recommendation 4 — Use Education and Experience Segmentation for Workforce Planning

Build separate hiring strategies for entry, mid, and senior levels. Align campus recruiting with roles where education shows the strongest payoff, and design senior hiring pipelines for high-experience, high-salary roles in AI-heavy domains.

Recommendation 5 — Expand Remote-Enabled Hiring for Strong-Outcome Roles

Where remote ratio correlates with better salary and growth outcomes, broaden geographic sourcing to access talent beyond local markets, reduce shortage risk, and improve compensation competitiveness without proportionally increasing fixed costs.

11. Impact Estimation, Limitations & Future Scope

11.1 Impact Estimation

The following estimates are derived from the project's KPIs and regression outputs:

- Wage Premium Estimate: High AI impact roles show a measurable average salary difference compared to low AI impact roles. After regression controls for experience, education, and industry, the AI-specific salary premium is quantified by the AI Impact coefficient in the salary regression model.
- Vulnerability Estimate: High automation risk roles show meaningfully lower average growth rates and are disproportionately represented among declining roles — supporting automation-driven displacement as a real labor market dynamic.
- Industry Prioritization: Top growth industries concentrate a large share of 'increasing' roles while representing a smaller share of total roles, indicating that demand is concentrated and investment should be targeted accordingly.

11.2 Limitations

- Forecast-based growth rates include uncertainty; results depend on projection assumptions and may not capture macroeconomic shocks.
- AI impact and automation risk may be proxy scores; measurement definitions can vary across data sources.
- Regression models assume linear/additive relationships; some effects may be non-linear.
- Observational data supports association, not strict causality.
- Dataset coverage may not represent all geographies or informal employment sectors.

11.3 Future Scope

- Add time-series labor data (multi-year observed trends) to validate projections against realized outcomes.
- Include geography and firm size variables to better explain salary and growth differences at a granular level.
- Apply non-linear models (gradient boosting / random forest) to capture interaction effects such as AI impact × industry.
- Add skill-level text analytics using job description NLP for deeper explainability of demand signals.
- Build scenario analysis covering optimistic, base, and pessimistic automation adoption trajectories.